Leonardo Scaramelli

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Data Science MSc graduate with a research background in Applied Probability, MCMC methods, Bayesian Statistics and Diffusion Models.

Education

Bocconi University, MSc Data Science / Statistics — **Milan, Italy**
Relevant Classes: Bayesian Statistics, Dynamic Optimization, Algorithms, Machine Learning, Probability Theory, Computer Vision, Stochastic Calculus — Sept. 2022 – Apr. 2025
Thesis: "Theoretical Convergence Guarantees of Diffusion Models", advisor: Prof. Giacomo Zanella

Bocconi University, BSc Economics and Finance — **Milan, Italy**
Relevant Classes: Accounting, Securities Markets, Monetary Economics, Financial Mathematics, Applied Stochastic Processes, Introduction to Options and Futures — Sept. 2019 – Sept. 2022
Thesis: "Stochastic Modeling and Simulation-Based Approach for Derivatives Pricing", advisor: Prof. Giacomo Zanella

University of Luxembourg, Erasmus Major in Economics and Mathematics — **Luxembourg, LU**
Relevant Classes: Corporate Finance, Economic Growth, Linear Algebra II, Dynamical Systems — Jan. 2022 – Jun. 2022

Experience

Data Science Intern, ION Group — **Milan, Italy**
Computer Vision, RAG, LLM — *May 2023 – Aug. 2023*

- Developed a computer-vision pipeline for facial feature extraction and real-time face detection/recognition from ID images using FaceNet, integrated into an LLM-based chatbot.
- Delivered an end-to-end MVP and presented the solution to the client's management team alongside ION's Founder.

Visiting Student, Bocconi Institute for Data Science and Analytics — **Milan, Italy**
Bayesian Statistics and Diffusion Models, Mentor: Prof.ssa Sonia Petrone — *Apr. 2023 – Sept. 2025*

- Studied Bayesian nonparametric (BNP) methods, starting with Dirichlet processes and Pòlya tree priors; presented research papers to faculty and peers.
- Studied the mathematical foundations of diffusion models and their applications; presented research on Schrödinger Bridges for accelerating diffusion-based sampling in Bayesian computation. Implemented the paper.
- Member of Bayesian Learning Laboratory, research unit with focus on Bayesian Statistics and Machine Learning.

Projects

Research Project, UCL | *Applied Probability, supervised by Prof. Samuel Livingstone* — **Dec. 2025 – Jun. 2026**

- Investigated step-size randomization in MCMC algorithms using Weak Poincaré Inequalities to characterize convergence behavior.
- Investigated theoretically and numerically the existence of drift conditions for MALA targeting distributions with lighter-than-Gaussian tails.

Self-Driving Cars Project | *Python, TensorFlow/Keras* — **Nov. 2023 – Dec. 2023**

- Developed and evaluated deep-learning models for 20-class semantic segmentation and drivable-area detection on the BDD100K dataset, using mIoU as the primary evaluation metric.
- Compared custom U-Net/FCNN architectures with transfer-learning approaches using ResNet, VGG16, MobileNetV2 and DenseNet121 encoders, investigating data augmentation, fine-tuning and architectural choices.
- Developed and evaluated segmentation models on a custom BDD100K split, achieving 36.52% semantic-segmentation mIoU and 71.42% drivable-area mIoU.

Technical Skills

Languages: Python (proficient), R (familiar), SQL
ML & Statistics: PyTorch, TensorFlow, scikit-learn, Hugging Face Transformers
Data & Cloud: Numpy, Pandas, PySpark, Databricks, AWS
Tools: Git, LangChain